

Assessing Human-LLM Alignment in the Evaluation of Odia Spelling and Grammatical Errors

HSL7702: COMPUTATIONAL LINGUISTICS (NATURAL LANGUAGE UNDERSTANDING)

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Evaluating LLMs for Indic languages is as much a linguistic problem as a computational one. Standard benchmarks rarely capture the structural precision that native speakers rely on (Watts et al., 2024).

- **The Odia case.** Odia and Bengali show the largest human-LLM agreement gaps in the PARIK-SHA study, even when global alignment looks reasonable (Watts et al., 2024). Broad fluency scores mask failures in fine-grained morpho-syntax.
- **From quality assessment to linguistic diagnostics.** Most evaluation work asks *what* a model says. This study asks *how accurately* it detects and corrects structural errors, tested against native human intuition across five grammatical error types.
- **The missing gold standard.** There is currently no established benchmark to verify whether an AI judge truly understands Odia morpho-syntactic integrity or is simply pattern-matching surface form.

- **BEA-2019 Shared Task** (Bryant et al., 2019). Established that unsupervised pre-training on synthetic data transfers well to low-resource grammatical error correction, motivating the study of GEC in Indic languages.
- **MuCGEC** (Zhang et al., 2022). Multi-reference evaluation with character- and subword-level metrics improves robustness for complex scripts, an important consideration for Odia.
- **DeCoGLM** (Li & Wang, 2024). A localised detection-correction architecture achieving near-SOTA GEC with faster inference; its error-detection framing directly informs our C1 evaluation dimension.
- **LLMs for Chinese GEC** (Lin, 2024). LLMs offer strong zero-shot baselines but struggle with fine-grained errors such as word choice, a pattern we test for in Odia morphology and case-marking.
- **LLMCL-GEC** (Fang et al., 2024). Curriculum learning (simple to complex) yields consistent gains in GEC, suggesting that structured prompting strategies matter for low-resource settings.
- **ReaLMistake** (Kamoi et al., 2024). LLMs can label grammatical errors but frequently fail to reason about them correctly. This is the direct motivation behind evaluating span quality and linguistic description separately (our C2 dimension).

Core question

To what extent can automated LLM evaluation reliably replace human linguistic expertise for Odia grammatical error correction, and where precisely does alignment break down?

RQ	Question
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|------------|---|
| RQ1 | To what extent does a frontier LLM's judgement align with the nuanced intuition of a human expert when correcting Odia errors? |
| RQ2 | Can AI reliably detect the full spectrum of Odia structural failures, from orthographic noise (spelling, script normalization) to morpho-syntactic errors (agreement, <i>Vibhakti</i> /case-marking, tense/aspect) and code-mixing? |
| RQ3 | Does the AI judge exhibit self-preference bias or favour verbosity over linguistic accuracy when scoring its own error analyses? |
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Corpus construction

- **Source.** Odia translations of the BHASHA 2025 IndicGEC dataset (Bhattacharyya et al., 2025), containing carefully **injected** errors, and the documented C4 Corpus (Dodge et al., 2021), containing **naturally occurring** language-specific errors.
- **Size.** 50 sentences, balanced at 10 per category, fully human-audited before use.
- **Annotator LLMs.** Sarvam-105B and LLaMA-4-Scout-17B-16E-Instruct each processed all 50 sentences independently, producing 100 annotation instances in total.

Five gold error categories

- 1 Script Normalization
- 2 Spelling & Typographical Error
- 3 Grammatical Error (morpho-syntactic: agreement, case-marking, tense/aspect)
- 4 Code-Mixing / Wrong Language
- 5 Correct Sentence (no error present)

Zero-shot prompting

Both annotator LLMs were given a fixed system prompt instructing them to act as Odia linguistic judges with no examples. Output was constrained to a strict JSON schema capturing:

- `error_exists` (binary)
- `error_span` (Error location)
- `category_type` (taxonomy label)
- `proposed_correction` (surface form)
- `linguistic_analysis` (reasoning)

Four evaluation dimensions

Six evaluators (2 human, 4 LLM) scored every annotation:

- C1** Error Detection (0/1)
- C2** Span + Description (0/1/2)
- C3** Error Category (0/1)
- C4** Corrected Sentence (0/1/2)

Evaluators: Human 1, Human 2, Sarvam-105B (self), LLaMA (self), GPT-Oss-120B, Gemini-3.1-Flash-Lite-Preview.

Agreement Metrics: Formulas and Scope

Metric	Formula	Scope and note
Raw % agreement	$PA = \frac{\text{agreeing pairs}}{\text{total pairs}}$	Exact and adjacent (≤ 1 step) variants; no chance correction
Cohen's κ	$\kappa = \frac{p_o - p_e}{1 - p_e}$	Pairwise; linear-weighted variant for ordinal C2/C4 (Cohen, 1960, 1968)
Fleiss' κ	$\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$	Multi-rater nominal; pooled marginals (Fleiss, 1971)
Gwet's AC1/AC2	$AC = \frac{p_o - p_e^{(w)}}{1 - p_e^{(w)}}$	Robust to prevalence skew; AC2 adds ordinal weights (Gwet, 2008)
Krippendorff's α	$\alpha = 1 - \frac{D_o}{D_e}$	Multi-rater, any scale; ordinal variant uses rank-weighted δ (Krippendorff, 1970)
ICC(A,1) / ICC(C,1)	$\frac{BMS - EMS}{BMS + (k-1)EMS}$ (A,1 adds $k(JMS - EMS)/n$)	Absolute vs. consistency reliability for C2/C4 (Shrout & Fleiss, 1979)
Spearman's ρ / Kendall's τ	$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}; \tau = \frac{2(n_c - n_d)}{n(n-1)}$	Rank correlation for all rater pairs; ρ measures monotonic association, τ concordant vs. discordant pairs (Spearman, 1904; Kendall, 1938)

Rater Groups (G1 to G7)

Group	Composition
G1	Human 1 + Human 2
G2	Sarvam-105B + LLaMA (<i>annotator</i>)
G3	GPT-Oss-120B + Gemini (<i>non-annotator</i>)
G4	All 4 LLMs
G5	Humans + Annotator LLMs
G6	Humans + Non-annotator LLMs
G7	All 6 raters

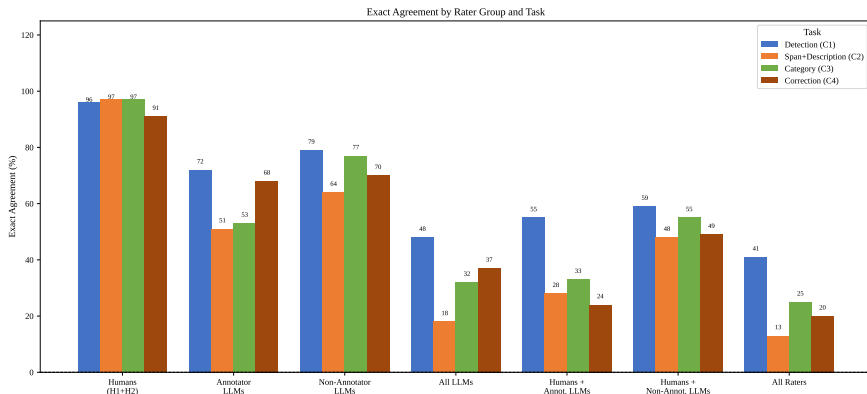
The self-evaluation distinction

G2 (annotator LLMs) produced the GEC outputs being judged. When they also score those outputs, a conflict of interest arises: the same model is both annotator and judge. This is the condition under which self-preference (optimism) bias is expected (Panickssery, Bowman, & Feng, 2024).

G3 (non-annotator LLMs) have no stake in the output. They serve as neutral third-party evaluators.

Results

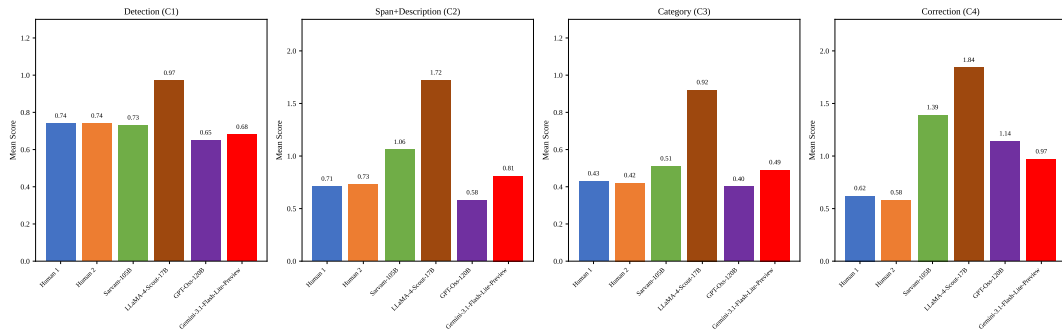
Raw Percentage Agreement by Rater Group



- **G1 (humans):** 91 to 97% exact agreement across all tasks, well above every LLM group.
- **G2 (annotator LLMs):** 51 to 72%, the lowest of any two-rater group; adjacent agreement on C2 rises to 75%, indicating that extreme disagreements between Sarvam and LLaMA are uncommon.
- As group size grows toward G7, unanimous agreement drops to around 41% on C1 and 13% on C2, which is expected when six diverse systems must all agree.

Mean Scores per Rater: Leniency Bias

Mean Score per Rater per Task



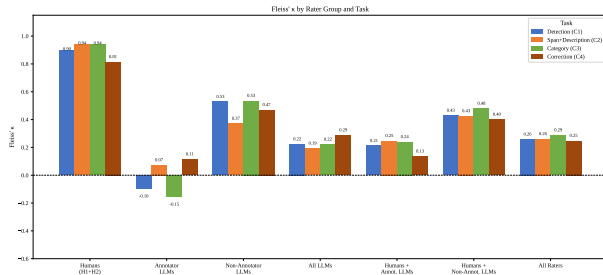
- LLaMA-4-Scout assigns the highest mean scores by a wide margin: 1.72 on C2 and 1.84 on C4, versus roughly 0.8 to 1.1 for all other raters.
- This systematic leniency is a direct source of low agreement within G2.
- Human 1 and Human 2 show nearly identical mean profiles across all four tasks, which is consistent with their near-perfect exact agreement.

Pairwise Cohen's κ : All Evaluator Pairs

Pair	C1 Detection	C2 Span+Desc.	C3 Category	C4 Correction
H1 vs H2	0.90	0.94	0.94	0.81
H1 vs Sarvam	0.15	0.24	0.28	0.17
H1 vs LLaMA	0.02	0.14	0.09	0.06
H1 vs GPT	0.28	0.15	0.20	0.31
H1 vs Gemini	0.32	0.48	0.48	0.36
H2 vs Sarvam	0.10	0.26	0.30	0.13
H2 vs LLaMA	0.02	0.13	0.08	0.05
H2 vs GPT	0.23	0.17	0.21	0.24
H2 vs Gemini	0.37	0.50	0.50	0.33
Sarvam vs LLaMA	0.01	0.16	0.04	0.17
Sarvam vs GPT	0.35	0.34	0.42	0.42
Sarvam vs Gemini	0.35	0.35	0.44	0.45
LLaMA vs GPT	0.05	0.09	0.08	0.14
LLaMA vs Gemini	0.00	0.12	0.08	0.17
GPT vs Gemini	0.53	0.38	0.54	0.48

- The human pair reaches $\kappa = 0.90$ to 0.94 across all tasks, placing them in the “almost perfect” band (Landis & Koch, 1977).
- LLaMA vs. Sarvam-105B: $\kappa = 0.01$ on C1 and sub-chance on C3, indicating their detection and category judgments are essentially unrelated to each other.
- Gemini-3.1-Flash-Lite-Preview comes closest to humans among the LLMs, reaching $\kappa \approx 0.32$ to 0.50 depending on task.

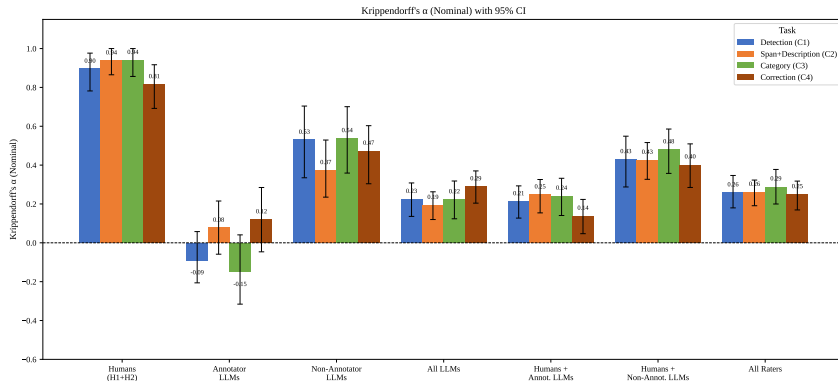
Fleiss' κ and Gwet's AC₁ by Rater Group



- G2 (annotator LLMs): Fleiss' $\kappa_F = -0.098$ on C1 and -0.153 on C3. Agreement below chance reflects systematically opposing judgements between Sarvam and LLaMA, not mere random noise.

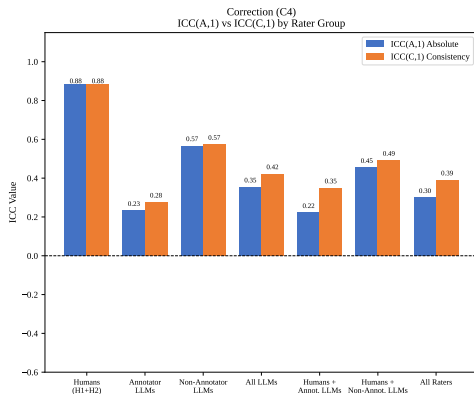
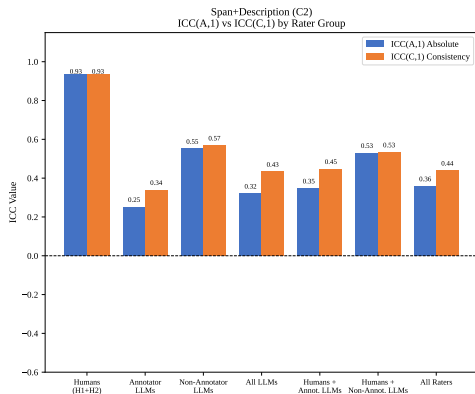
- **Kappa paradox ruled out.** Gwet's AC₁ equals Fleiss' κ exactly on all binary tasks (e.g. G2 C1: both = -0.098 ; G1 C3: both = 0.939). Identical values confirm the low G2 scores are substantive, not a prevalence artefact. For ordinal tasks, AC₂ rises above AC₁ (e.g. G3 on C2: $0.371 \rightarrow 0.468$), showing partial-credit consistency exceeds exact agreement.
- **G6 (Humans + GPT + Gemini)** reaches Fleiss' $\kappa = 0.43$ on C3 and AC₁ = 0.48 , the best results among any multi-rater group. Including either annotator LLM in any group reduces agreement monotonically.

Krippendorff's α : Primary Multi-Rater Metric



- G1 achieves $\alpha \geq 0.81$ on all tasks with narrow 95% bootstrap CIs (Efron, 1979), confirming the two humans as a reliable benchmark.
- G2: $\alpha = -0.093$ (C1) and -0.148 (C3); CIs spanning zero mean these values are statistically indistinguishable from chance agreement.
- G6 is the strongest mixed group at $\alpha = 0.43$ and 0.48 on C1 and C3 respectively. Ordinal α consistently exceeds nominal α across all groups (e.g. G3 on C2: $0.37 \rightarrow 0.56$), so when LLMs disagree, they are rarely more than one step apart.

ICC(A,1) vs. ICC(C,1): Absolute Agreement vs. Consistency



- G1: $ICC(A,1) = ICC(C,1) = 0.933$ on C2 and 0.884 on C4; the negligible gap confirms no directional bias between the two humans (Shrout & Fleiss, 1979; Koo & Li, 2016).
- G2: $ICC(C,1) > ICC(A,1)$ on both tasks. The annotator LLMs rank items in a broadly similar order but apply systematically different absolute score thresholds. The gap between the two ICC values directly measures this scale-offset.
- G6 reaches moderate reliability (0.53 on C2), the best for any mixed group.

Unit of analysis

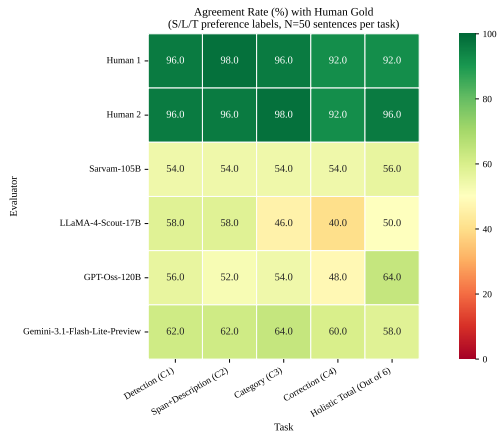
Each of the 50 sentences is evaluated by all 6 raters on each task. For every sentence $\times$ task combination, each rater is assigned a **preference label**: **S** (Sarvam scores higher), **L** (LLaMA scores higher), or **T** (Tie). This transforms the data into a 3-class nominal categorisation problem ($N = 50$ sentences, 6 raters) — the minimum required for valid chance-corrected statistics.

Why not Kendall's τ ? Rank correlation requires a ranking of ≥ 3 items. With only two systems (Sarvam vs. LLaMA), any two-item list trivially yields $\tau = \pm 1$, making the statistic undefined. Rescaling an agreement rate as $2r - 1$ is mathematically equivalent to a percentage, not a correlation.

Valid metrics applied:

- **Raw agreement rate** with human gold S/L/T label (per evaluator, task, category)
- **Optimism bias**: self-evaluator margin minus human gold margin
- **Multi-rater**: Fleiss' κ , Krippendorff's α , Gwet's AC_1 on the 50-sentence label matrix
- **Pairwise**: Cohen's κ (6×6)

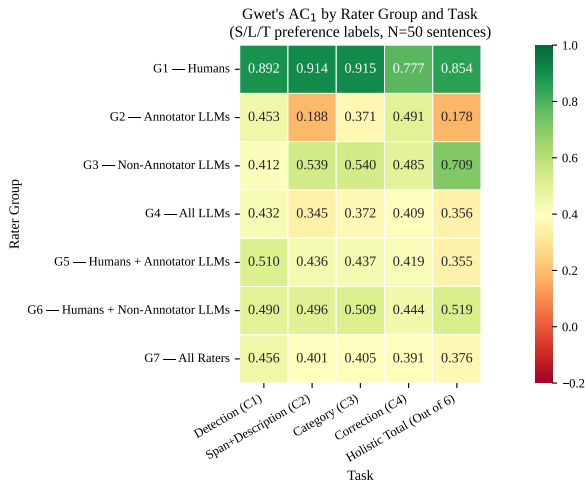
System Preference: Agreement with Humans



Agreement rate with human S/L/T label ($N = 50$ per task)

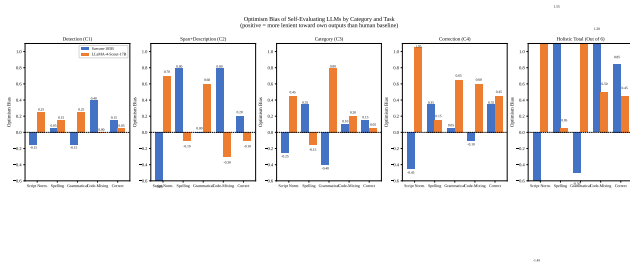
- Human 1 and Human 2: 92–98% across all tasks, confirming the gold label reliability.
- Gemini: **60–64%** (best LLM evaluator)
- GPT-Oss-120B: 48–56%
- Sarvam-105B: 54% uniformly across tasks
- LLaMA: **40–58%**; self-preference bias systematically deflates agreement with the human gold preference label.

Multi-Rater Chance-Corrected Agreement on S/L/T Labels



- **G1 (humans):** Gwet's $AC_1 \geq 0.78$ on all tasks, confirming the human gold label is reliable.
- **G2 (annotator LLMs):** $AC_1 = 0.19$ on C2 — the lowest group-task cell in the table — driven by Sarvam and LLaMA systematically preferring opposite systems.
- **G3 (non-annotator LLMs):** $AC_1 = 0.41$ – 0.54 , consistently outperforming G2 and confirming that conflict-of-interest distorts annotator LLM judgements.
- G6 (Humans + Non-Annotator LLMs) is the best mixed group on C2 and C3.

Optimism Bias of Self-Evaluating LLMs



- Optimism bias is defined here as the difference between a self-evaluator's score margin for its own outputs and the corresponding human gold margin. A positive value means the LLM rates itself more favourably than humans do.
- LLaMA shows strong positive bias on every task: +0.58 on C4, +0.27 on C3, +0.16 on C2, +0.14 on C1.
- Sarvam-105B is substantially better calibrated, with near-zero bias on C1 and C3, though moderate on C2 (+0.24).
- These patterns align with the self-preference findings reported by Panickssery, Bowman, & Feng (2024).

Summary and Conclusions

What held across all seven metrics

- **G1 (Humans):** $\kappa \geq 0.81$, $\text{ICC} \geq 0.88$ on all tasks; narrow bootstrap CIs confirm stability as a reliable benchmark.
- **Gemini-3.1-Flash-Lite-Preview** is the closest LLM to humans on every agreement measure.
- **G2 (Annotator LLMs)** falls below chance on error detection and category assignment ($\kappa_F < 0$, $\alpha < 0$), driven chiefly by LLaMA's leniency bias.

Patterns worth noting

- Ordinal metrics consistently exceed their nominal counterparts; disagreements are rarely more than one scale step apart.
- Non-annotator LLMs (GPT-Oss-120B, Gemini) are more reliable evaluators than the self-evaluating annotator LLMs.
- C2 and C4 show the widest spread across rater groups and are the most sensitive tasks to evaluator type.

Error Category Consistency Across Rater Groups

Composite = mean of all valid IAA metrics across four tasks. Range $[-1, +1]$; higher = more consistent.

Category	G1 H-H	G7 All
Code-Mixing	0.962	0.392
Spelling	0.902	0.310
Grammatical	0.885	0.265
Script Norm.	0.854	0.170
Correct	0.839	0.300

Category	G2 Ann.	G6 H+NonAnn.
Code-Mixing	0.140	0.529
Spelling	0.065	0.540
Grammatical	0.030	0.457
Correct	-0.039	0.332
Script Norm.	-0.043	0.216

Key patterns

- **Code-Mixing** ranks 1st or 2nd in all six groups. Non-Odia-script tokens are unambiguous to any evaluator.
- **Script Normalization** is hardest for humans but easiest for LLM-only groups, which detect character variants via pattern matching. Scores are not comparable across evaluator types.
- **Grammatical Error** ranks 3rd–4th across all groups. Identifying the error type (C3) is easier than correcting it (C4), which demands morpho-syntactic knowledge.
- **G2 falls below chance** on Script Norm. and Correct Sentence; self-preference outweighs the linguistic signal in these categories.

Limitations and Scope Constraints

Dataset and Statistical Power

- **Corpus Size:** The evaluation relies on a carefully curated but minimal corpus of 50 sentences. This restricts statistical power and limits broad generalizability across all Odia domains.
- **Human Rater Pool:** The “gold standard” consensus is derived from only two expert human evaluators, which may under-represent acceptable linguistic variations in ambiguous grammatical cases.

Methodological Constraints

- **Zero-Shot Prompting:** Annotator LLMs were evaluated using a strict zero-shot protocol. This isolates baseline capabilities but may underestimate true performance achievable with few-shot examples or rubric anchoring.
- **Correction Ambiguity:** Complex morpho-syntactic errors often have multiple valid corrections. The framework can severely penalise models that propose correct but structurally divergent outputs from the human baseline.
- **Model Scale Variability:** The annotator LLMs were not constrained to identical parameter scales; selecting models with comparable parameter sizes could enable more controlled comparative evaluation.

Scaling the study

- Expand the number of sentences per category; add Noise and Multi-error categories with three or more human annotators for majority-vote conflict resolution.

Pairwise evaluation and Elo-based ranking

- Implement the pairwise comparison track (battles, stratified by category) using Standard Elo and MLE Elo (Bradley-Terry model) to rank annotator LLMs.
- Apply pairwise Kendall's τ between human and LLM Elo leaderboards; test whether the DA-pairwise gap in PARIKSHA (Watts et al., 2024) replicates for Odia GEC.
- Investigate rubric-anchored prompting and score normalisation to reduce the scale-offset bias seen in G2.

- ❶ Bhattacharyya, P., N J, K., Terdalkar, H., Jagadeeshan, M. B., Nigam, S. K., Susmitha, A. S., & Bhattacharya, A. (2025). Findings of the IndicGEC and IndicWG shared task at BHASHA 2025. In A. Bhattacharya et al. (Eds.), *Proceedings of the 1st Workshop on Benchmarks, Harmonization, Annotation, and Standardization for Human-Centric AI in Indian Languages* (BHASHA 2025) (pp. 127–134). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.bhasha-1.12>
- ❷ Bryant, C., Felice, M., Andersen, Ø. E., & Briscoe, T. (2019). The BEA-2019 shared task on grammatical error correction. In *Proceedings of the Fourteenth Workshop on Innovative Use of NLP for Building Educational Applications* (pp. 52–75). Association for Computational Linguistics. <https://doi.org/10.18653/v1/W19-4406>
- ❸ Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46. <https://doi.org/10.1177/001316446002000104>
- ❹ Cohen, J. (1968). Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4), 213–220. <https://doi.org/10.1037/h0026256>
- ❺ Dodge, J., Sap, M., Marasovic, A., Agnew, W., Ilharco, G., Groeneveld, D., & Gardner, M. (2021). Documenting the English Colossal Clean Crawled Corpus. *arXiv preprint arXiv:2104.08758*. <https://arxiv.org/abs/2104.08758>
- ❻ Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. <https://doi.org/10.1214/aos/1176344552>

- 7 Fang, T., Wong, D. F., Zhang, L., Jin, K., Zhang, Q., Li, T., Hou, J., & Chao, L. S. (2024). LLMCL-GEC: Advancing grammatical error correction with LLM-driven curriculum learning. *arXiv preprint arXiv:2412.12541*.
- 8 Fleiss, J. L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5), 378–382. <https://doi.org/10.1037/h0031619>
- 9 Gwet, K. L. (2008). Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1), 29–48. <https://doi.org/10.1348/000711006X126600>
- 10 Gwet, K. L. (2014). *Handbook of inter-rater reliability* (4th ed.). Advanced Analytics.
- 11 Hayes, A. F., & Krippendorff, K. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures*, 1(1), 77–89. <https://doi.org/10.1080/19312450709336664>
- 12 Kamoi, R., Das, S. S., Lou, R., Ahn, J. J., Zhao, Y., Lu, X., Zhang, N., Zhang, Y., Zhang, R. H., Vummanthala, S. R., Dave, S., Qin, S., Cohan, A., Yin, W., & Zhang, R. (2024). Evaluating LLMs at detecting errors in LLM responses. *arXiv preprint arXiv:2404.03602*. <https://doi.org/10.48550/arXiv.2404.03602>
- 13 Kendall, M. G. (1938). A new measure of rank correlation. *Biometrika*, 30(1–2), 81–93. <https://doi.org/10.1093/biomet/30.1-2.81>
- 14 Koo, T. K., & Li, M. Y. (2016). A guideline for selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15(2), 155–163. <https://doi.org/10.1016/j.jcm.2016.02.012>

- 15 Krippendorff, K. (1970). Estimating the reliability, systematic error, and random error of interval data. *Educational and Psychological Measurement*, 30(1), 61–70. <https://doi.org/10.1177/001316447003000105>
- 16 Krippendorff, K. (2011). *Computing Krippendorff's alpha-reliability*. Departmental Papers (ASC), University of Pennsylvania.
- 17 Krippendorff, K. (2013). *Content analysis: An introduction to its methodology* (3rd ed.). SAGE Publications.
- 18 Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174. <https://doi.org/10.2307/2529310>
- 19 Li, W., & Wang, H. (2024). Detection-correction structure via general language model for GEC. *arXiv preprint arXiv:2405.17804*.
- 20 Lin, S. (2024). Evaluating LLMs' GEC performance in learner Chinese. *PLOS ONE*, 19(10), e0312881. <https://doi.org/10.1371/journal.pone.0312881>
- 21 Panickssery, A., Bowman, S. R., & Feng, S. (2024). LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, 37. <https://doi.org/10.48550/arXiv.2404.13076>
- 22 Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2), 420–428. <https://doi.org/10.1037/0033-2909.86.2.420>
- 23 Spearman, C. (1904). The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1), 72–101. <https://doi.org/10.2307/1412159>

- 24 Watts, I., Gumma, V., Yadavalli, A., Seshadri, V., Swaminathan, M., & Sitaram, S. (2024). PARIKSHA: A large-scale investigation of human-LLM evaluator agreement on multilingual and multi-cultural data. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics.
- 25 Zhang, Y., Li, Z., Bao, Z., Li, J., Zhang, B., Li, C., Huang, F., & Zhang, M. (2022). MuCGEC: A multi-reference multi-source evaluation dataset for Chinese grammatical error correction. In M. Carpuat, M. de Marneffe, & I. V. Meza Ruiz (Eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 3118–3130). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2022.naacl-main.227>